# Lecture 7

# Techniques of Differentiation

**1. Constant**

$$f(x) = c$$
$$f'(x) = \frac{d}{dx}(c) = 0$$

**2. Power Functions**

$$f(x) = x^n$$
$$f'(x) = \frac{d}{dx}(x^n) = nx^{n-1}$$

**3. Constant × Power Function**

$$f(x) = cx^n$$
$$f'(x) = c \cdot nx^{n-1}$$

**4. Sums / Differences**

$$\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}\big(f(x)\big) \pm \frac{d}{dx}\big(g(x)\big)$$

**5. Product Rule**

$$f(x) = u(x) \cdot v(x)$$
$$\frac{df}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

**6. Quotient Rule**

$$f(x) = \frac{u(x)}{v(x)}$$
$$\frac{df}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

**7. Trigonometric Functions**

$$\frac{d}{dx}(\sin x) = \cos x, \quad \frac{d}{dx}(\cos x) = -\sin x, \quad \frac{d}{dx}(\tan x) = \sec^2 x$$
$$\frac{d}{dx}(\sec x) = \sec x \tan x, \quad \frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x, \quad \frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$$

**8. Chain Rule**

If $y = f\big(g(x)\big)$, Let, $g(x) = u$

Then, $\frac{dy}{dx} \quad = \frac{df}{du} \cdot \frac{du}{dx}$

**OR (Alternative)**

$$\frac{d}{dx}\big[f\big(g(x)\big)\big] \quad = f'\big(g(x)\big) \cdot g'(x)$$

**OR (General Form)**

$$\frac{d}{dx}[f(u)] \quad = f'(u) \cdot \frac{du}{dx}$$

**9. Exponential and Logarithmic Functions**

$$y \quad = \ln x \quad , \quad \frac{dy}{dx} \quad = \frac{1}{x}$$
$$y \quad = \log_a x, \quad \frac{dy}{dx} \quad = \frac{1}{x\ln a}$$
$$y \quad = e^x \quad , \quad \frac{dy}{dx} \quad = e^x$$
$$y \quad = u^v \quad , \quad \frac{dy}{dx} \quad = u^v \frac{d}{dx}(vlnu)$$

**10. Inverse Trigonometric Functions**

$$\frac{d}{dx}(\sin^{-1}x) \quad = \frac{1}{\sqrt{1-x^2}}, \qquad \frac{d}{dx}(\cos^{-1}x) \quad = -\frac{1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx}(\tan^{-1}x) \quad = \frac{1}{1+x^2}, \qquad \frac{d}{dx}(\cot^{-1}x) \quad = -\frac{1}{1+x^2}$$
$$\frac{d}{dx}(\sec^{-1}x) \quad = \frac{1}{|x|\sqrt{x^2-1}}, \quad \frac{d}{dx}(\mathrm{cosec}^{-1}x) \quad = -\frac{1}{|x|\sqrt{x^2-1}}$$

**<u>Higher Derivatives:</u>**

$$f'(x) \quad = \frac{df}{dx}$$
$$f''(x) \quad = \frac{d^2f}{dx^2}$$
$$f'''(x) \quad = \frac{d^3f}{dx^3}$$
$$\vdots$$

The n-th derivative of $f(x)$ is, $f^{(n)}(x) = \frac{d^nf}{dx^n}$

General n-th Order Derivative at a specific point $x = a$,

$$f^{(n)}(a) = \left.\frac{d^n f}{dx^n}\right|_{x=a}$$

**<u>Successive Differentiation:</u>**

Continuously differentiating the results of differentiation.

**Notation**

1st derivative: $f'(x)$ or $\frac{dy}{dx}$ or $D_x y$ or $y_1$

2nd derivative: $f''(x)$ or $\frac{d^2y}{dx^2}$ or $D_x^2 y$ or $y_2$

3rd derivative: $f'''(x)$ or $\frac{d^3y}{dx^3}$ or $D_x^3 y$ or $y_3$

$\vdots$

n-th derivative: $f^{(n)}(x)$ or $\frac{d^ny}{dx^n}$ or $D_x^n y$ or $y_n$

**Example:** Find the n-th Derivative of $y = (ax + b)^n$. (Successive Differentiation)

**Solution:**

Given,

$$\begin{aligned}
y &= (ax+b)^n \\
\frac{dy}{dx} &= n(ax+b)^{n-1} \cdot a = na(ax+b)^{n-1} \\
\frac{d^2y}{dx^2} &= na(n-1)(ax+b)^{n-2} \cdot a = n(n-1)a^2(ax+b)^{n-2} \\
\frac{d^3y}{dx^3} &= n(n-1)a^2(n-2)(ax+b)^{n-3} \cdot a = n(n-1)(n-2)a^3(ax+b)^{n-3} \\
&\vdots \\
\frac{d^ny}{dx^n} &= n(n-1)(n-2)\dots(n-(n-1))a^n(ax+b)^{n-n} \\
&= n(n-1)(n-2)\dots 1\, a^n(ax+b)^0 \\
&= n!\ a^n
\end{aligned}$$

$\therefore \frac{d^ny}{dx^n} = n!\ a^n$, is the n$^{\text{th}}$ derivative of $y$.

**Example:** Find the n-th Derivative of $y = \ln(ax + b)$.

**Solution:** Given, $y = \ln(ax + b)$

$$\frac{dy}{dx} = \frac{a}{ax+b} = 0!\, a(ax+b)^{-1}$$

$$\frac{d^2y}{dx^2} = a \cdot (-1)(ax+b)^{-1-1} \cdot a = -a^2(ax+b)^{-2} = -\frac{1!\, a^2}{(ax+b)^2}$$

$$\frac{d^3y}{dx^3} = -a^2(-2)(ax+b)^{-3} \cdot a = 2a^3(ax+b)^{-3} = \frac{2!\, a^3}{(ax+b)^3}$$

$$\frac{d^4y}{dx^4} = 2a^3(-3)(ax+b)^{-4} \cdot a = -6a^4(ax+b)^{-4} = -\frac{3!\, a^4}{(ax+b)^4}$$

$$\frac{d^5y}{dx^5} = -6a^4(-4)(ax+b)^{-5} \cdot a = 24a^5(ax+b)^{-5} = \frac{4!\, a^5}{(ax+b)^5}$$

$$\vdots$$

$$\frac{d^ny}{dx^n} = (-1)^{n+1} \frac{(n-1)!a^n}{(ax+b)^n}$$

So, $\frac{d^ny}{dx^n} = (-1)^{n+1} \frac{(n-1)!a^n}{(ax+b)^n}$, is the n-th derivative of y.

**Example:** Find the n-th Derivative of $y = \sin(ax + b)$.

**Solution:** Given, $y = \sin(ax + b)$

$$\frac{dy}{dx} = \cos(ax+b) \cdot a = a\cos(ax+b)$$

$$\frac{d^2y}{dx^2} = -\sin(ax+b) \cdot a^2 = -a^2\sin(ax+b)$$

$$\frac{d^3y}{dx^3} = -\cos(ax+b) \cdot a^3 = -a^3\cos(ax+b)$$

$$\frac{d^4y}{dx^4} = \sin(ax+b) \cdot a^4 = a^4\sin(ax+b)$$

$$\frac{d^5y}{dx^5} = \cos(ax+b) \cdot a^5 = a^5\cos(ax+b)$$

$$\vdots$$

$$\frac{d^ny}{dx^n} = (-1)^{\left\lfloor \frac{n}{2} \right\rfloor} a^n \begin{cases} \sin(ax+b) & \text{if } n \text{ is even} \\ \cos(ax+b) & \text{if } n \text{ is odd} \end{cases}$$

Where, $\left\lfloor \frac{n}{2} \right\rfloor$ is the floor function.

**Floor Function:**

$\left\lfloor \frac{1}{2} \right\rfloor = 0$

$\left\lfloor \frac{3}{2} \right\rfloor = 1$

$\left\lfloor \frac{5}{2} \right\rfloor = 2$

**Ceiling Function:**

$\left\lceil \frac{1}{2} \right\rceil = 1$

$\left\lceil \frac{3}{2} \right\rceil = 2$

$\left\lceil \frac{5}{2} \right\rceil = 3$

## H.W. Practice Sheet a(1–7)

**Example:** If $y = e^{ax}\sin bx$, then show that $y_2 - 2ay_1 + (a^2 + b^2)y = 0$.

**Solution:** Given, $y = e^{ax}\sin bx$

$$\begin{aligned}
\text{So, } y_1 &= \frac{d}{dx}(e^{ax}\sin bx) \\
&= e^{ax}\frac{d}{dx}(\sin bx) + \sin bx\frac{d}{dx}(e^{ax}) \\
&= e^{ax}(\cos bx)b + \sin bx(e^{ax})a \\
&= e^{ax}b\cos bx + e^{ax}a\sin bx
\end{aligned}$$

$$\begin{aligned}
\text{So, } y_2 &= \frac{d}{dx}(e^{ax}b\cos bx + e^{ax}a\sin bx) \\
&= e^{ax}b(-\sin bx)b + e^{ax}b \cdot a \cdot \cos bx + e^{ax}a \cdot \cos bx \cdot b + e^{ax} \cdot a \cdot a \cdot \sin bx \\
&= -e^{ax}b^2\sin bx + e^{ax}ab\cos bx + e^{ax}ab\cos bx + e^{ax}a^2\sin bx \\
&= (a^2 - b^2)e^{ax}\sin bx + 2abe^{ax}\cos bx
\end{aligned}$$

$$\begin{aligned}
\text{So, } y_2 - 2ay_1 + (a^2 + b^2)y &= (a^2 - b^2)e^{ax}\sin bx + 2abe^{ax}\cos bx - 2a(e^{ax}b\cos bx + e^{ax}a\sin bx) \\
&\quad +(a^2 + b^2)e^{ax}\sin bx \\
&= (a^2 - b^2 + a^2 + b^2)e^{ax}\sin bx + 2abe^{ax}\cos bx - 2abe^{ax}\cos bx - 2a^2e^{ax}\sin bx \\
&= 2a^2e^{ax}\sin bx - 2a^2e^{ax}\sin bx \\
&= 0
\end{aligned}$$

So, $y_2 - 2ay_1 + (a^2 + b^2)y = 0$ [Showed]

## H.W. Practice Sheet (C)

**H.W.** If $y = \ln\left(x + \sqrt{x^2 + 1}\right)$, show that $(1 + x^2)y_2 + xy = 0$.

**Example:** $y = \ln\cos x$. Find $\frac{dy}{dx}$. (Chain rule)

**Solution:** Given, $y = \ln\cos x$.

$$\begin{aligned}
\text{So, } \frac{dy}{dx} &= \frac{d}{dx}(\ln\cos x) \\
&= \frac{1}{\cos x} \cdot (-\sin x) \\
&= -\frac{\sin x}{\cos x} \\
&= -\tan x
\end{aligned}$$

**H.W.** Find $\frac{dy}{dx}$ for the following functions:

i. $y = e^{x^2}$

ii. $y = x^2\cos\frac{1}{x}$ (chain rule+product rule) Ans: $2x\cos\left(\frac{1}{x}\right) + \sin\left(\frac{1}{x}\right)$

## H.W. Practice Sheet 1 (i-iii) (product rule)

**Example:** Find the Differential Coefficient of the function $y = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$, with respect to $x$. (Inverse Trigonometric Function)

**Solution:** Given, $y = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

$$= \cos^{-1}\left(\frac{1-\tan^2\theta}{1+\tan^2\theta}\right)$$

$$= \cos^{-1}(\cos 2\theta)$$

$$= 2\theta$$

$$= 2\tan^{-1}x$$

Let, $x = \tan\theta$.

So, $\theta = \tan^{-1}x$

**Formulas:**

$$\tan 2A = \frac{2\tan A}{1-\tan^2 A}$$

$$\sin 2A = \frac{2\tan A}{1+\tan^2 A}$$

$$\cos 2A = \frac{1-\tan^2 A}{1+\tan^2 A}$$

So, $\frac{dy}{dx} = \frac{d}{dx}(2\tan^{-1}x) = 2 \cdot \frac{1}{1+x^2} = \frac{2}{1+x^2}$.

**Example:** Find the Differential Coefficient of the function $y = \tan(\sin^{-1}x)$, with respect to $x$.

**Solution:** Given, $y = \tan(\sin^{-1}x)$.

Let, $x = \sin\theta$.

So, $\theta = \sin^{-1}x$

and, $\cos\theta = \sqrt{1-x^2}$

Now, $y = \tan(\sin^{-1}x) = \tan\theta$.

So, $$\frac{dy}{dx} = \frac{d}{dx}(\tan\theta)$$

$$= \sec^2\theta \cdot \frac{d\theta}{dx}$$

$$= \frac{1}{\cos^2\theta} \cdot \frac{d}{dx}(\sin^{-1}x)$$

$$= \frac{1}{\cos^2\theta} \cdot \frac{1}{\sqrt{1-x^2}}$$

$$= \frac{1}{1-x^2} \cdot \frac{1}{\sqrt{1-x^2}}$$

$$= \frac{1}{(1-x^2)^{\frac{3}{2}}}$$

**Example:** Find the Differential Coefficient of the function $y = \cot^{-1}\left(\frac{1+x}{1-x}\right)$, with respect to $x$.

**Solution:** Given, $y = \cot^{-1}\left(\frac{1+x}{1-x}\right)$

Let, $x = \tan\theta$.

So, $\theta = \tan^{-1} x$

Now, $y = \cot^{-1}\left(\frac{1+x}{1-x}\right)$

$$= \cot^{-1}\left(\frac{1+\tan\theta}{1-\tan\theta}\right)$$

$$= \cot^{-1}\left(\frac{\tan\frac{\pi}{4}+\tan\theta}{1-\tan\frac{\pi}{4}\cdot\tan\theta}\right) \qquad [\text{ as } \tan\frac{\pi}{4} = 1]$$

$$= \cot^{-1}(\tan(\frac{\pi}{4}+\theta)) \qquad \left[\text{as } \tan(A+B) = \frac{\tan A+\tan B}{1-\tan A\tan B}\right]$$

$$= \cot^{-1}(\cot\{\frac{\pi}{2} - (\frac{\pi}{4}+\theta)\})$$

$$= \frac{\pi}{2} - (\frac{\pi}{4}+\theta)$$

$$= \frac{\pi}{4} - \theta$$

So, $\frac{dy}{dx} = \frac{d}{dx}\left(\frac{\pi}{4} - \theta\right)$

$$= 0 - \frac{d\theta}{dx}$$

$$= -\frac{d}{dx}(\tan^{-1} x)$$

$$= -\frac{1}{1+x^2}$$

## H.W. Practice Sheet 1 (iv–x)

**Implicit Differentiation:**

$y$ is not given as a function of $x$ explicitly.

**Example:** Find $\frac{dy}{dx}$ at the point $(1, 2)$ on the curve $3x^4 - x^2y + 2y^3 = 0$.

**Solution:** Given, $3x^4 - x^2y + 2y^3 = 0$

Differentiating both sides with respect to $x$,

$$\frac{d}{dx}(3x^4 - x^2y + 2y^3) = 0$$

$\Rightarrow \frac{d}{dx}(3x^4) - \frac{d}{dx}(x^2y) + \frac{d}{dx}(2y^3) = 0$

$\Rightarrow 12x^3 - x^2\frac{dy}{dx} - 2xy + 6y^2\frac{dy}{dx} = 0$

$\Rightarrow (6y^2 - x^2)\frac{dy}{dx} = 2xy - 12x^3$

$\Rightarrow \frac{dy}{dx} = \frac{2xy - 12x^3}{6y^2 - x^2}$

At the point (1, 2),

$$\left.\frac{dy}{dx}\right|_{(1,2)} = \frac{2(1)(2) - 12(1)^3}{6(2)^2 - (1)^2}$$
$$= \frac{4 - 12}{24 - 1}$$
$$= \frac{-8}{23}$$

So, $\left.\frac{dy}{dx}\right|_{(1,2)} = -\frac{8}{23}$.

## H.W. Practice Sheet 4(ii)

**Example:** $x^y = y^x$. Find $\frac{dy}{dx}$. (Logarithmic and Implicit Differentiation)

**Solution:** Given, $x^y = y^x$

$\Rightarrow \ln(x^y) = \ln(y^x)$

$\Rightarrow y\ln x = x\ln y$

Differentiating both sides with respect to $x$,

$\frac{d}{dx}(y\ln x) = \frac{d}{dx}(x\ln y)$

$\Rightarrow \frac{dy}{dx}\ln x + y \cdot \frac{1}{x} = \ln y + x \cdot \frac{1}{y}\frac{dy}{dx}$

$\Rightarrow \frac{dy}{dx}\ln x - \frac{x}{y} \cdot \frac{dy}{dx} = \ln y - \frac{y}{x}$

$\Rightarrow \frac{dy}{dx}\left(\ln x - \frac{x}{y}\right) = \ln y - \frac{y}{x}$

When you have the variable as power, then you should take ln on the both sides of equation or in the function before differentiating. Just be careful, if taking the ln leads to an undefined value (ln(x) is only defined for x>0), you can't use it.

So, $\frac{dy}{dx} = \frac{\ln y - \frac{y}{x}}{\ln x - \frac{x}{y}}$

**H.W.** Find $\frac{dy}{dx}$ for the following functions:

i. $\cos(x+y) = e^{x+y}$ [ Hint: take ln on both side of the equation] Ans: -1.

ii. $xy + e^y = 0$ [Hint: Can you really take ln on the both sides here? Treat it as normal implicit function] Ans: $\frac{-y}{x+e^y}$

## (Only Logarithmic)

**Example:** $y = x^2 e^x sinxcoshx$. Find $\frac{dy}{dx}$.

**Solution:** Given, $y = x^2 e^x sinxcoshx$.

$$\Rightarrow \ln y = \ln(x^2 e^x sinxcoshx)$$

$$\Rightarrow \ln y = \ln(x^2) + \ln(e^x) + \ln(\sin x) + \ln(\cosh x) \quad [\text{as } ln(xy) = lnx + lny]$$

$$\Rightarrow \ln y = 2\ln x + x\ln(e) + \ln(\sin x) + \ln(\cosh x)$$

$$\Rightarrow \ln y = 2\ln x + x + \ln(\sin x) + \ln(\cosh x) \quad [\text{ as } \ln(e) = 1]$$

Differentiating both sides with respect to $x$,

$\frac{d}{dx}(\ln y) = \frac{d}{dx}(2\ln x + x + \ln(\sin x) + \ln(\cosh x))$

$$\Rightarrow \frac{1}{y}\frac{dy}{dx} = \frac{2}{x} + 1 + \frac{\cos x}{\sin x} + \frac{1}{\cosh x}\sin hx$$

$$\Rightarrow \frac{dy}{dx} = y\left(\frac{2}{x} + 1 + \frac{\cos x}{\sin x} + \frac{1}{\cosh x}\sin hx\right)$$

So, $\frac{dy}{dx} = x^2 e^x sinxcoshx\left(\frac{2}{x} + 1 + \cot x + \tan hx\right)$.

**H.W.** Find $\frac{dy}{dx}$ for the following functions:

i. $y = x^{\sin x}$ Ans: $x^{\sin x}(\frac{\sin x}{x} + \ln x \cdot \cos x)$

ii. $y = a^x$ Ans: $a^x \cdot \ln(a)$

## H.W. Practice Sheet 2(i)

**Example:** $y = (\sin x)^{\cos x} + (\cos x)^{\sin x}$. Find $\frac{dy}{dx}$.

**Solution:** Given, $y = (\sin x)^{\cos x} + (\cos x)^{\sin x}$

Let, $z = (\sin x)^{\cos x}, \quad p = (\cos x)^{\sin x}$

So, $y = z + p$

Now, $z = (\sin x)^{\cos x}$

$\Rightarrow \ln z = \cos x \ln(\sin x)$

$$\Rightarrow \frac{d}{dx}(\ln z) = \frac{d}{dx}\big(\cos x \ln(\sin x)\big)$$

$$\Rightarrow \frac{1}{z}\frac{dz}{dx} = \left[-\sin x \ln(\sin x) + \cos x \cdot \frac{1}{\sin x} \cdot \cos x\right]$$

$$\Rightarrow \frac{dz}{dx} = z[-\sin x \ln(\sin x) + \cot x \cdot \cos x]$$

So, $\frac{dz}{dx} = (\sin x)^{\cos x}[-\sin x \ln(\sin x) + \cot x \cdot \cos x]$

Again, $p = (\cos x)^{\sin x}$

$\Rightarrow \ln p = \sin x \ln(\cos x)$

$$\Rightarrow \frac{d}{dx}(\ln p) = \frac{d}{dx}\big(\sin x \ln(\cos x)\big)$$

$$\Rightarrow \frac{1}{p}\frac{dp}{dx} = \cos x \ln(\cos x) + \sin x \cdot \left(\frac{-\sin x}{\cos x}\right)$$

$$\Rightarrow \frac{dp}{dx} = p[\cos x \ln(\cos x) - \sin x \tan x]$$

So, $\frac{dp}{dx} = (\cos x)^{\sin x}[\cos x \ln(\cos x) - \sin x \tan x]$

Now, $y = z + p$

$$\Rightarrow \frac{dy}{dx} = \frac{dz}{dx} + \frac{dp}{dx}$$

So, $\frac{dy}{dx} = (\sin x)^{\cos x}[-\sin x \ln(\sin x) + \cot x \cdot \cos x] + (\cos x)^{\sin x}[\cos x \ln(\cos x) - \sin x \tan x]$.